Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience in **QA-oriented software delivery** across **Python**, **Java**, **C++**, **SQL**, and Linux. Strong in test design, automation, debugging, CI/CD, regression validation, and release support for reliable production systems.

Education

University of Southern California **Los Angeles, CA**
M.S. in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Implemented **FastAPI** validation and release workflows for **5K+ monthly jobs**, adding reconciliation checks, incident debugging support, and reliability safeguards to sustain **99%+ uptime**.
- Built backend services on **Azure SQL**, **PostgreSQL**, and cloud storage for a 500+ user platform, improving latency from **1.2s to 350ms** through schema, indexing, and query tuning.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** platform handling **10K+ submissions**, with secure ingestion paths, modular APIs, and instructor-facing workflows backed by **MySQL/AWS RDS**.
- Implemented concurrent processing and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while improving operational stability and reducing turnaround time by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python experimentation and simulation pipelines for autonomous systems, processing **100K+ trajectory logs** and improving long-horizon planning performance by **18%**.
- Documented model behavior and evaluation results while collaborating on stable training, debugging, and closed-loop validation workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with microservice architecture and **20+ unit/integration suites** using **Jenkins**, PyTest, and Mockito across **3 operating systems** in an Agile environment.
- Troubleshot API and ETL issues using **OOP patterns**, **AWS Glue**, and S3, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** platform with **REST APIs**, **WebSockets**, and microservices, supporting **8+ workflows** and **2M+ synchronized records** in a user-facing operations system.
- Automated deployments and validation with **Jenkins**, Docker, and Azure SQL-backed services, reducing deployment time by **50%** while reaching **95%+ successful build validation**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL
QA & Validation: Test Design, Regression, Functional & Performance Testing, Bug Reproduction, Root Cause Analysis
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP
Systems & Tooling: Linux, Ubuntu, Git, Maven, Docker, Jenkins, GitHub Actions, JIRA
Cloud & DevOps: AWS (RDS, S3, Glue), Microsoft Azure, Kubernetes, CI/CD, DevOps
Engineering Practices: Unit Testing, Integration Testing, Debugging, Documentation, Agile, System Design

Projects

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented **6 scheduling algorithms** with interactive controls and live visualizations across **1K+ simulated process combinations**, strengthening systems reasoning, debugging, and validation-focused tooling.

Chat2SQL | *Java, TypeScript, SQL, Maven, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, accelerating query authoring, schema exploration, structured execution, and developer testing across a **Java + TypeScript** codebase.

HMIS | *MERN, TypeScript, ABDM APIs, Twilio*
- Designed a workflow platform with structured schemas, API integrations, and operational monitoring supporting **100+ daily appointments** while cutting record retrieval time by **2x**.